ShelfSense Progress Update

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ShelfSense (CSE499A, Section 10, Group 5) has finished Sprints 1-3, processed 115 new supermarket videos, and trained its Sprint 4 product detector (mAP50 0.913 on SKU-110K). Reading brands on detected product crops doubles the brands the VLM finds compared with whole frames.

What to show today

Show the evidence trail from raw video to detected products, about 10 minutes in this order. All files are in `E:\ShelfSense`.

1. **The problem (1 min).** Brand companies pay shops for fridge space and placement, but field auditors can be bribed. ShelfSense replaces the auditor's word with video evidence and an automatic audit.
2. **The design (1 min).** Hybrid pipeline: keyframe extraction, then YOLO detects every product, then a small VLM reads brands on crops, then rules compute the KPIs. The VLM alone cannot count (Sprint 3 result).
3. **Keyframe extraction, live (2 min).** Run `python -m src.extraction.extract_frames --video Test_Videos\VID-20261018-WA0092.mp4 --fps 4`, then open its `frames_log.csv` (kept / blurry / duplicate per frame). Show `data\results\figures\blur_check.jpg`: sharp frames kept, smeared ones dropped.
4. **The new footage (1 min).** `data\results\figures\videos_overview_1.jpg` and `_2.jpg`: one frame from each of our supermarket videos.
5. **Baseline VLM failures (2 min).** Sprint 3 table below: the VLM invents brands and answers "10" for every count.
6. **The detector (3 min).** In `data\results\figures`: `sku110k_training_curves.png` (mAP rising over 30 epochs), `prelabel_examples.jpg` (boxes on 12 of our test frames), then `detector_good_and_bad.jpg` to show where it works and where it fails, and crop_brand_examples.jpg for brand reading on crops. Optional live run: `python -m src.detection.prelabel --weights runs\detect\sku110k_v8n\weights\best.pt` (about 1 min).
7. **Tests (30 s).** `python -m pytest -q`: 39 automated tests pass.

Progress by sprint

Sprints 1-3 are done; Sprint 4′s detector is trained and waiting for corrected labels on our own frames.

Sprint	Focus	Status	Main result
1	Setup and problem definition	Done	Repo, environment, scope, KPIs (cooler purity, paid placement, out-of-stock, share of shelf)
2	Data and keyframe extraction	Extractor done; tested on 3 fridge clips + 115 supermarket videos	5,264 frames sampled, 4,558 kept; Crete blur score replaced Laplacian variance
3	Baseline VLM (Qwen2-VL-2B, zero-shot)	Done on 3 fridge clips	Brand precision 0.13-0.30; every count = 10 or 0; prompts fix format, not perception
4	Product detection (YOLOv8n)	Detector trained; 222 frames pre-labelled; crop brand test done	See the Sprint 4 section
5	Integration and rule engine	Not started	
6	Evaluation and demo app	Not started	

Sprint 3 in one line: two prompt versions cut unparseable answers from 24 to 8, but brands were still invented (Pepsi, 7up in a Coca-Cola fridge) and counts were never real. This is why Sprint 4 adds a detector to do the counting.

Latest update: 115 supermarket videos

We recorded 115 unique shelf videos (about 22 minutes) in a Bangladeshi supermarket and ran the full extraction on them.

- **Footage:** 119 files, of which 4 were exact copies (found by SHA-256 hash) and 4 were stock clips we excluded. The 111 own clips are 576x1024 portrait, median 11 s.

- **Content:** open shelves, not branded fridges: personal care, biscuits and snacks, noodles, tea and coffee, juices, soft-drink cans, milk, rice and lentils, oil. Price tags (TK) and "OFFER" labels are visible.
- **Split:** 82 train and 29 test videos. Neighbouring clips film the same shelf, so the split keeps blocks of consecutive videos together and the test set never overlaps training.
- **Blur filter, first real test:** the score ranks frames correctly (sharp 0.25-0.35, smeared 0.58-0.62). Near the 0.5 threshold there is no clean gap, so we kept 0.5.
- **Annotation set:** the sharpest 2 frames per video, 164 train and 58 test, chosen automatically.

Split	Videos	Duration (s)	Frames sampled	Kept	Blurry	Duplicate
Train	82	934	3,771	3,420	54	297
Test	29	322	1,298	1,071	115	112

These videos suit product detection, share of shelf and price tags. They cannot test cooler purity, which still needs branded-fridge recordings.

Sprint 4: product detector

YOLOv8n trained on 2,000 SKU-110K shelf photos finds products with **mAP50 0.913** (mAP50-95 0.564) on 500 held-out SKU-110K photos.

Setting (500 SKU-110K val images)	mAP50	mAP50-95	Precision	Recall
1024 px, up to 1,000 boxes (chosen)	0.913	0.564	0.897	0.844
1024 px, up to 299 boxes	0.889	0.549	0.897	0.833
640 px (default size), up to 1,000 boxes	0.866	0.510	0.875	0.792

- **Training:** one class (`product` , no brands), 30 epochs, 163.5 min on a laptop GTX 1650 (4 GB). mAP was still rising at epoch 30, so more epochs or images should help.
- **Why the settings matter:** one SKU-110K photo holds 718 products and 11 of 500 hold more than 300, so the default 300-box limit costs 2.4 mAP50 points. Small products need 1024 px: 640 px costs 4.7 points.
- **On our 115 videos:** 151,279 products found on 4,558 keyframes, median 29 per frame.
- **Works:** packed shelves at normal distance (toothpaste shelf 63 boxes, drinks shelf 35, nearly every item).

- **Fails:** big products close to the camera. Milk-carton close-ups get a median of 6 boxes per frame, against 53 for canned food. SKU-110K products are small, so the model has not seen close-ups.
- **Not yet measured:** mAP on our own frames. That needs corrected labels; the model has pre-labelled all 222 annotation frames (median 30 boxes each) for correction in Roboflow.

Figures in `data\results\figures`: `sku110k_training_curves.png`, `prelabel_examples.jpg` (12 test frames), `detector_good_and_bad.jpg` (two good shelves, two close-up failures).

Brand reading: single-product crops vs whole frames

Asking the VLM about one detected product at a time doubles the brands it finds (recall 0.63 vs 0.30) at precision 0.94. This supports the hybrid design chosen after Sprint 3.

Setup: Qwen2-VL-2B, fp16, zero-shot (same model as Sprint 3). 12 frames from the 6 drinks-shelf test videos; 120 random detector crops. Answers are matched to a catalogue of the 24 expected brands, as an audit for a brand company would.

Method (12 frames)	Usable answers	Brand precision	Brand recall
Whole frame, "list the brands"	6 of 12 (6 looped until the token limit)	1.00 (on 6 frames)	0.30
Detector crops, "which brand is this?"	12 of 12	0.94	0.63

- **Per crop:** 48 of 67 readable crops named correctly (72%; 67% before catalogue matching). Without the catalogue, many answers are near-misses ("chabac", "canad dry", "sprit").
- **Wrong brands are rare but biased:** 2 of 120 crops got a different real brand after matching, both "Coca-Cola" (for Canada Dry and Sprite). This is the Sprint 3 pull towards famous brands, so violations still need a human check of the evidence crop.
- **No answer:** 11 readable crops, mostly Thai or Bengali script (Doi Kham, Latina) and vertical text.
- **Logos:** on crops with no legible text the VLM still named Gatorade (from its G logo) and Fanta correctly.

Caveats: crop and frame labels were drafted by Claude and need a team check; 6 videos show patterns, not significance; the catalogue was built from the test labels, so it is an upper bound. On this 4 GB laptop each crop takes 21 s (model partly in CPU memory).

Figure: `data\results\figures\crop_brand_examples.jpg`.

Next steps

1. Correct the detector's pre-labels on our 222 frames in Roboflow and draw empty gaps. Two team members check each test frame, since these are the ground truth.
2. Fine-tune the detector on our 164 train frames and report mAP and counting error on our 58 test frames (script ready: `src/detection/finetune_ours.py`).
3. Verify the crop and frame brand labels (team check), then build the per-client brand catalogue into the pipeline and test crops from fridge videos for cooler purity.
4. Record branded beverage fridges (15-30 s each, original quality), including some with competitor products, to test cooler purity.
5. Sprint 5: connect extraction, detection and VLM into one pipeline that writes the JSON audit report.

Open issues: the Sprint 3 ground truth is still a draft (fridge_03 brands unverified). The video categories were labelled from one frame each and need a team check. SKU-110K's license allows academic use only, so its photos are never shared.